GeoMEx-VLM: A Geo-Memory Guided Sparse Expert Vision–Language Model for Multigranularity Asset Grounding in Remote Sensing

Hessa Alfraihi, Umar Raza Mukhtar, and Hamza Mukhtar

***Abstract*—Remote sensing (RS) vision–language models (VLMs) have advanced instruction-following interpretation for captioning, VQA, dialog, and grounding. However, fine-grained remote sensing interpretation remains challenging because objects are often small, densely distributed, rotated, occluded, or visually similar to surrounding infrastructure. Existing RS-VLMs mainly rely on dense shared reasoning, coarse spatial outputs, or limited box-level grounding, restricting their ability to jointly support scene understanding, counting, axis-aligned bounding box (AAB) grounding, rotation-aware bounding box (RAB) localization, and mask prediction. To address these limitations, we propose a geo-memory-guided sparse expert VLM (GeoMEx-VLM) for multigranularity RS interpretation and strategic asset grounding. GeoMEx-VLM introduces a class-scale geo-memory (CSGM) module to retrieve class- and scale-aware geospatial priors and inject them into visual token modeling and sparse expert routing. A geo-memory-guided router assigns tokens to experts using token features, task embeddings, and retrieved geo-context, enabling class-aware and task-adaptive reasoning. Unified spatial tokenization further represents AABs, RABs, and compressed masks as language-compatible outputs within a single autoregressive decoder. Experiments demonstrate high performance on multigranularity RS interpretation, achieving 82.36 Pr@0.5 on DIOR-RSVG language-conditioned horizontal grounding and 75.42 mIoU on referring segmentation, with strong performance on rotation-aware localization, visual question answering, and strategic asset counting. These results validate the effectiveness of geo-memory-guided sparse reasoning and unified spatial tokenization for precise strategic asset understanding. The code is available at https://github.com/umeraza/remote_sensing**

***Index Terms*—Axis-aligned box grounding, class-scale geo-memory (CSGM), referring expression segmentation, remote sensing (RS), rotation-aware localization, sparse mixture-of-experts.**

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I. Introduction

REMOTE sensing (RS) imagery supports Earth observation, urban monitoring, transportation analysis, environmental assessment, and strategic asset understanding. With increasing availability of high-resolution aerial and satellite imagery [18], [19], RS has expanded from scene classification to VQA, captioning, counting, visual grounding, oriented localization, and referring expression segmentation [2]. These tasks remain challenging because target objects are often small, dense, rotated, occluded, or visually similar to surrounding infrastructure. Recent RS-vision–language models (VLMs), including GeoChat [1], RS-LLaVA [2], RSGPT [3], SkyEyeGPT [4], EarthGPT [5], EarthDial [6], and EarthGPT-X [7], have improved instruction-following RS understanding. Recent work also highlights a shift toward data-centric adaptation of general-purpose VLMs with diverse tasks and domain-specific supervision [44]. However, most existing models still rely on dense shared reasoning and coarse spatial outputs, limiting unified scene understanding, counting, rotation-aware grounding, and mask-level prediction.

Spatial precision remains a major challenge in RS vision–language grounding because aerial imagery contains objects with large-scale variation, arbitrary orientation, dense distributions, and weak visual contrast. DOTA [12] shows the limitations of horizontal boxes for representing oriented aerial objects, while FAIR1M [19] highlights fine-grained recognition challenges in high-resolution imagery. Recent RS-VLMs [13], [14] extend grounding beyond coarse coordinates. GeoGround [13] unifies horizontal boxes, oriented boxes, and masks as textual outputs [13], while GeoPix [14] and GeoPixel [15] support pixel-level grounding. However, existing methods still provide limited integration of multiformat grounding with class-aware spatial priors and task-specialized reasoning.

Current RS-VLMs make limited use of recurring class-wise geospatial regularities, although RS scenes often exhibit stable spatial and contextual patterns [17]. Geo-context prototypes and spatial priors have improved region-aware representation and reduced class confusion in pretraining and segmentation [16], [17], [18] but remain underexplored for decoder-only RS-VLMs, multiformat grounding, and expert routing. Sparse mixture-of-experts learning [40] addresses task heterogeneity through specialized reasoning pathways. Existing sparse RS-VLMs, including RS-MoE [8], RSUniVLM [9], RingMoE

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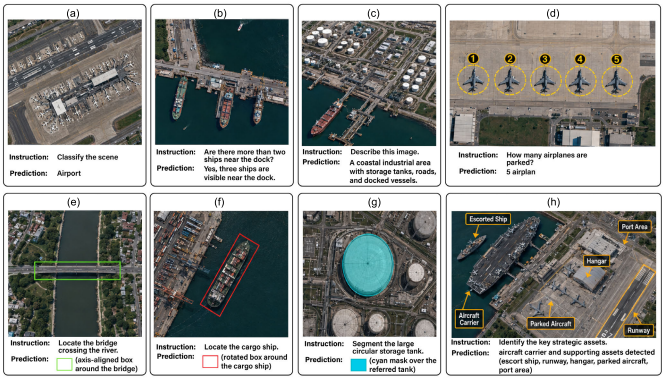


Fig. 1. Qualitative overview of GeoMEx-VLM across multitask RS interpretation. (a) Scene classification. (b) VQA. (c) Image captioning. (d) Object counting. (e) AAB grounding. (f) RAB grounding. (g) Expression segmentation. (h) Strategic asset grounding and understanding.

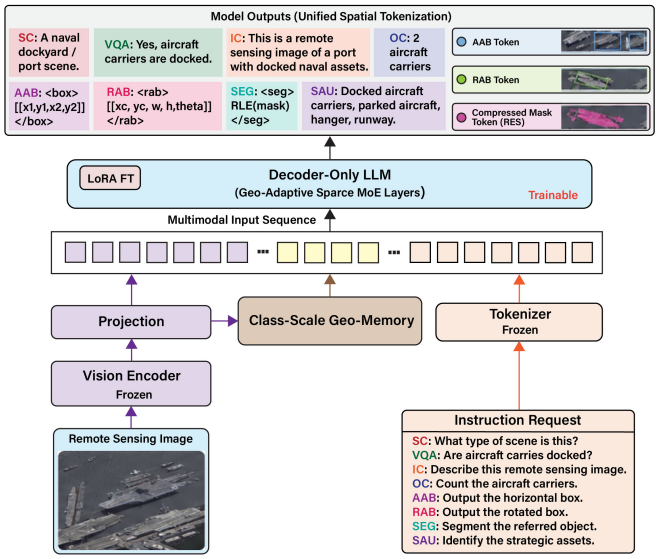


Fig. 2. Architecture of GeoMEx-VLM. Visual, textual, and CSGM tokens are processed by a decoder-only LLM with LoRA and geo-adaptive sparse MoE layers to generate unified semantic and spatial outputs.

[10], and SkyMoE [11], route experts using task, modality, instruction, or granularity cues, but make limited use of class-conditioned geospatial memory for expert assignment.

To address these limitations, we propose a geo-memory-guided sparse expert vision–language framework (GeoMEx-VLM) for multigranularity remote sensing interpretation and fine-grained asset grounding. GeoMEx-VLM introduces a class-scale geo-memory (CSGM) module that stores and retrieves class- and scale-aware geospatial priors for recurring RS objects, injecting them into both visual token modeling and sparse expert routing instead of only refining visual features as in conventional memory mechanisms [17], [18]. The framework further uses unified spatial tokenization to represent axis-aligned boxes, rotation-aware boxes, and compressed masks as language-compatible tokens, enabling natural-language responses and multiformat spatial predictions within one autoregressive model. Training follows a progressive dense-to-sparse strategy. Stage 1 establishes dense remote sensing visual–language alignment through instruction tuning, while Stage 2 converts selected FFN layers into sparse MoE layers and activates geo-memory-guided routing for task- and granularity-specific expert specialization. The final objective combines autoregressive generation, expert load balancing, geo-memory alignment, and cross-format grounding supervision, avoiding excessive task-specific heads while supporting scene classification, VQA, captioning, counting, horizontal grounding, oriented grounding, segmentation, and multiformat asset localization and remote sensing interpretation within a unified architecture, as shown in Fig. 1.

The main contributions of this work are summarized as follows.

1) We propose a geo-memory-guided sparse expert VLM for multigranularity remote sensing interpretation and fine-grained asset grounding across semantic, box-level, oriented, and mask-level outputs.
2) We introduce a CSGM module that stores class- and scale-aware geospatial priors and injects them into both visual token modeling and expert routing.
3) We design a geo-memory-guided expert routing layer that assigns tokens to experts using token features, task embeddings, and retrieved CSGM, enabling class-aware and task-adaptive sparse reasoning.
4) We develop unified spatial tokenization to represent horizontal boxes, oriented boxes, and compressed masks as language-compatible outputs for autoregressive multiformat grounding.
5) We formulate a compact training objective and a cross-representation spatial agreement diagnostic to improve and evaluate consistency among box-, oriented-box-, and mask-level predictions.

II. Proposed Model

GeoMEx-VLM is a sparse expert vision–language framework for multigranularity RS interpretation, combining a CSGM module for class-conditioned geospatial memory modeling with unified spatial tokenization for representing horizontal boxes, oriented boxes, and segmentation masks as language-compatible tokens. Given an RGB remote sensing image $I \in \mathbb{R}^{H\times W\times 3}$ and an instruction query Q, the image is first converted into the evaluation view or spatial tiles required by the fixed-resolution visual backbone. The pretrained visual encoder then extracts patch-level visual tokens from each 336×336 encoder input, which are projected into the LLM hidden space and concatenated with the textual instruction embeddings. Class-conditioned geo-memory features are retrieved from the geo-context memory bank and injected into both visual representation learning and sparse expert routing to improve object-level discrimination. The decoder-only LLM then generates task-specific outputs, including natural-language answers, object counts, axis-aligned bounding box (AAB) tokens, rotation-aware bounding box (RAB) tokens, and compressed mask tokens, enabling scene classification, VQA, captioning, counting, horizontal grounding, oriented grounding, referring segmentation, and strategic asset understanding within a unified architecture, as shown in Fig. 2.

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A. Architecture of GeoMEx-VLM

Unlike dense VLMs, GeoMEx-VLM replaces selected FFN blocks with geo-adaptive sparse expert layers. Each expert is initialized from the corresponding pretrained Vicuna FFN parameters, which remain unchanged during Stage 1, and the resulting expert copies are subsequently optimized during Stage 2. A geo-memory-conditioned router assigns tokens to the resulting experts based on multimodal representations and retrieved geo-context. The visual encoder extracts patch-level image features, the projection layer maps them into the LLM embedding space, and textual instructions are encoded through the LLM word embeddings before joint multimodal processing. This architecture enables task- and granularity-specific sparse reasoning for scene understanding, object counting, visual grounding, segmentation, and strategic asset grounding while preserving the original Vicuna FFN parameters during Stage 1.

GeoMEx-VLM is constructed following the modular MLLM paradigm [27] of unimodal pretrained components. The visual encoder is instantiated from the CLIP ViT-L/14 model (24-layer Vision Transformer, 1024 hidden dimensions, 16 attention heads, patch size 14, and 336 × 336 input resolution). The 336 × 336 resolution denotes the input size of an individual encoder view and does not imply that a complete high-resolution remote sensing scene is globally compressed to this resolution. Large evaluation scenes are processed using deterministic spatial tiling before visual encoding. Each resulting tile is mapped to the 336 × 336 CLIP input, and spatial predictions are subsequently transformed back to the coordinate system of the original image. The decoder-only LLM is initialized from Vicuna-7B-v1.5 (32-layer Transformer, 4096 hidden dimensions, 32 attention heads, and 7B parameters). The learnable visual projection module, implemented as a single linear layer, maps image patch tokens into the LLM word-embedding space and is trained from scratch with Gaussian initialization ($\sigma = 0.02$).

1) Visual–Language Token Construction: Let $I \in \mathbb{R}^{H\times W\times 3}$ denote the input RS image. The visual encoder $E_v(\cdot)$ divides the image into patches and generates visual tokens $Z = E_v(I) = [z_1, z_2, \ldots, z_P] \in \mathbb{R}^{P\times C}$, where P is the number of visual patches and C is the feature dimension. A learnable projection module $f_p(\cdot)$ maps the visual sequence into the LLM hidden space as $V = f_p(Z) = [v_1, v_2, \ldots, v_P] \in \mathbb{R}^{P\times D}$, where D denotes the LLM embedding dimension. For the textual instruction Q, the tokenizer and word-embedding layer generate $T = g(Q) = [t_1, t_2, \ldots, t_N] \in \mathbb{R}^{N\times D}$, where N is the number of text tokens. The visual and textual tokens are concatenated into a unified multimodal sequence $X_0 = [v_1, v_2, \ldots, v_P, t_1, t_2, \ldots, t_N]$.

This multimodal sequence is passed through the decoder-only LLM. For compactness, let $\mathcal{A}(\cdot) = \mathrm{MSA}(\mathrm{LN}(\cdot))$ and $\mathcal{E}(\cdot) = \mathrm{MoE}(\mathrm{LN}(\cdot))$. The ℓth layer is updated as

$$X'_\ell = X_{\ell-1} + \mathcal{A}(X_{\ell-1}), \quad X_\ell = X'_\ell + \mathcal{E}(X'_\ell)$$
$$Y = \mathrm{LN}(X_L). \tag{1}$$

Here, MSA(·), LN(·), and MoE(·) denote multihead self-attention, layer normalization, and sparse expert transformation, respectively, and Y is the final representation after L LLM layers.

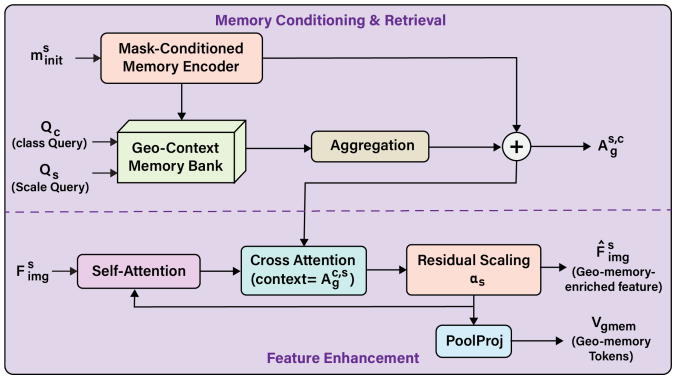


Fig. 3. CSGM module in GeoMEx-VLM. Training annotations supervise memory organization and spatial-prior learning, whereas inference retrieves geo-context using category, scale, and spatial queries predicted solely from the image–instruction pair.

2) CSGM Module: RS objects observed from nadir or near-nadir views often exhibit stable class-specific layouts and contextual patterns across scales. GeoMEx-VLM therefore introduces a CSGM module comprising a mask-conditioned memory encoder, a class-scale Geo-Memory Bank (GMB), and a geo-memory attention block, as shown in Fig. 3. The GMB is represented as $\mathcal{G} \in \mathbb{R}^{K\times L\times N_g\times D_g\times H_g\times W_g}$, where K, L, N_g, and D_g denote the numbers of classes, scales, memory slots, and memory dimensions, respectively, and $H_g \times W_g$ is the spatial resolution. Let $\mathcal{P}(\cdot) = \mathrm{PyramidProj}(\cdot)$. Memory retrieval and multiscale projection are defined as

$$H^{c,s}_g = \mathcal{G}(Q_c, Q_s), \qquad \{F^s_{img}\}^L_1 = \mathcal{P}\left(\{Z^s\}^L_1\right). \tag{2}$$

Here, $H^{c,s}_g$ denotes the retrieved class-scale memory and F^s_{img} denotes the projected visual feature at scale s. To prevent label leakage, category, scale, and spatial annotations are used only as training supervision. At validation and test time, Q_c and Q_s are inferred solely from the pre-memory image–instruction representations, and no ground-truth AAB, RAB, or mask is provided to CSGM. For localization tasks, m^s_{init} is estimated from query-conditioned attention or a coarse grounding prediction; for VQA and captioning, CSGM retrieves class-scale priors directly from multimodal image–instruction evidence.

The mask-conditioned encoder $I_g(\cdot)$ converts the initial mask into a spatial prior, while $\phi_g(\cdot)$ aggregates the retrieved class-scale memory. For compactness, let $\tilde{F}_s = \mathrm{SA}(F^s_{img})$ and $A_{c,s} = I_g(m^s_{init}) + \phi_g(H^{c,s}_g)$. The CSGM-enhanced visual feature is then

$$\hat{F}^s_{img} = \tilde{F}_s + \alpha_s \mathrm{CA}\left(\tilde{F}_s, A_{c,s}\right). \tag{3}$$

Here, SA(·) and CA(·) denote self-attention and cross-attention, respectively, and α_s is a learnable residual scaling factor. The enhanced multiscale features are pooled into geo-memory tokens, $V_{gmem} = \mathrm{PoolProj}(\{\hat{F}^s_{img}\}^L_{s=1})$, which are appended to the multimodal sequence as $X^*_0 = [V, V_{gmem}, T]$. This integration provides class- and scale-aware spatial priors for recognizing small, rotated, dense, and visually similar remote sensing objects.

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3) Task-Wise CSGM Inference: All reported validation and test results use annotation-free CSGM inference. Given only the image I and instruction Q, the pre-memory visual and textual representations are used to predict Q_c and Q_s; no ground-truth category, object scale, AAB, RAB, or mask is provided to the memory module, router, or decoder. Thus, evaluation follows the same input setting as a standard VLM and does not assume externally supplied semantic or spatial annotations.

The pre-memory visual and instruction tokens are pooled and projected into a joint multimodal representation $u_q = \psi_q(\text{Pool}([V, T]))$. The category and scale queries are then predicted as

$$(Q_c, Q_s) = \left(f_c\left(u_q\right),\ f_s\left(u_q\right)\right). \tag{4}$$

Here, $\psi_q(\cdot)$ denotes the multimodal projection, while $f_c(\cdot)$ and $f_s(\cdot)$ are learnable category- and scale-query projections, respectively. The resulting queries retrieve the relevant class-scale entries from the GMB without requiring an oracle class or object scale.

The inference behavior is task dependent. For VQA, Q_c and Q_s are inferred jointly from the question and image; explicit object mentions produce target-specific retrieval, whereas open-ended questions retain multiple relevant class-scale priors. For captioning, CSGM performs multiclass and multiscale retrieval because no single target is assumed. For counting, the queried object semantics guide Q_c, while visual evidence determines the relevant scale information. For AAB/RAB grounding and referring expression segmentation, the referring phrase provides target semantics, while query-conditioned attention or a coarse grounding prediction provides the spatial prior m^s_{init}. The retrieved geo-memory is then incorporated through (3), converted into V_{gmem}, and used by the geo-memory-guided sparse expert layers and autoregressive decoder.

Thus, the complete inference pipeline is $(I, Q) \rightarrow (V, T) \rightarrow (Q_c, Q_s, m_{init}) \rightarrow \text{CSGM} \rightarrow V_{gmem} \rightarrow \text{MoE} \rightarrow Y$.

4) Geo-Memory-Guided Expert Routing Layer: A standard MoE router assigns tokens to experts using only token-level hidden representations. This design is limited for RS interpretation because the same local pattern may require different reasoning depending on the queried class, object scale, spatial context, and task type. Therefore, we introduce a geo-memory-guided expert routing layer that conditions expert selection on token features, task information, and CSGM-derived geo-memory context.

Let $x'_{\ell,j} \in \mathbb{R}^D$ denote the jth token representation after the self-attention residual update at layer ℓ and $\bar{x}_{\ell,j} = [\text{LN}(X'_\ell)]_j$ denote its layer-normalized representation. The CSGM-enriched features are pooled into a compact geo-context vector g_c, while a task embedding e_τ represents the target operation, such as captioning, VQA, counting, AAB grounding, RAB grounding, segmentation, or strategic asset grounding. The geo-memory-conditioned routing distribution is computed as

$$\begin{aligned} g_c &= \psi_g\left(\text{Pool}\left(\left\{H^{c,s}_{ge}\right\}_{s=1}^{L}\right)\right) \\ r_{\ell,j} &= \text{Softmax}\left(W_r\left[\bar{x}_{\ell,j}; e_\tau; g_c\right]\right) \end{aligned} \tag{5}$$

where $\psi_g(\cdot)$ is a learnable projection layer, W_r is the router projection matrix, and $[\cdot;\cdot]$ denotes feature concatenation. The routing vector $r_{\ell,j} = [r^1_{\ell,j}, r^2_{\ell,j}, \ldots, r^E_{\ell,j}] \in \mathbb{R}^E$ defines a probability distribution over the E available experts, where $r^i_{\ell,j} \geq 0$ and $\sum_{i=1}^{E} r^i_{\ell,j} = 1$. Thus, each routing coefficient $r^i_{\ell,j}$ represents the relevance of expert i to token j at layer ℓ given its token content, target task, and retrieved geospatial context.

Rather than activating all experts for every token, only the k experts with the largest routing probabilities are selected. The sparse geo-adaptive expert transformation is formulated as

$$\begin{aligned} \mathcal{S}_{\ell,j} &= \text{TopKIdx}\left(r_{\ell,j}, k\right) \\ x_{\ell,j} &= x'_{\ell,j} + \sum_{i \in \mathcal{S}_{\ell,j}} r^i_{\ell,j} E_i\left(\bar{x}_{\ell,j}\right) \end{aligned} \tag{6}$$

where $\mathcal{S}_{\ell,j}$ denotes the indices of the k experts with the largest routing probabilities for token j at layer ℓ. $E_i(\cdot)$ denotes the ith expert, E is the total number of experts, and $k < E$ is the number activated per token. Each selected expert processes $\bar{x}_{\ell,j}$, and its output is weighted by the routing probability $r^i_{\ell,j}$ and added to $x'_{\ell,j}$ through a residual connection, yielding $X_\ell = [x_{\ell,j}]_{j=1}^{M}$. Because routing depends on token content, task identity, and CSGM-derived geo-context, (6) enables conditional sparse computation and task- and granularity-specific expert specialization for counting, RAB grounding, segmentation, and geospatial reasoning.

B. Unified Spatial Tokenization

GeoMEx-VLM converts geometric and pixel-level annotations into language-compatible spatial tokens, enabling natural-language responses and spatial predictions through the same autoregressive decoder. For horizontal grounding, an AAB $b_h = (x_1, y_1, x_2, y_2)$ is normalized, quantized to resolution R_h, and serialized as $t_{aab} =<$ `box` $>$ $[[\hat{x}_1, \hat{y}_1, \hat{x}_2, \hat{y} * 2]] <$ `/box` $>$. For oriented grounding, an RAB $b_o = (x_c, y_c, w, h, \theta)$ is encoded as $t * rab =<$ `rab` $>$ $[[\hat{x}_c, \hat{y}_c, \hat{w}, \hat{h}, \hat{\theta}]] <$ `/rab` $>$. These token formats preserve localization precision for small, elongated, and rotated remote sensing objects such as vehicles, aircraft, ships, bridges, runways, and storage tanks while keeping AAB, RAB, and compressed mask prediction within a unified generative framework.

1) Compressed Mask Tokens: For referring expression segmentation, the target mask is represented as a compact spatial-token sequence. Given a binary mask M, the mask is downsampled into a fixed grid $M_d = \text{Downsample}(M)$, where $M_d \in \{0, 1\}^{R_m \times R_m}$. Pixels belonging to the target object are encoded as 1, while background pixels are encoded as 0. To reduce sequence length, the binary mask is compressed using run-length encoding and written as $t_{mask} =<$ `seg` $>$ $\text{RLE}(M_d) <$ `/seg` $>$. These compressed mask tokens allow the model to approximate object shape without introducing a separate segmentation decoder. The representation also enables joint training of captioning, VQA, box grounding, oriented grounding, and segmentation under a unified language-generation objective.

C. Progressive Dense-to-Sparse Training

GeoMEx-VLM is trained using a progressive dense-to-sparse strategy, as shown in Fig. 4. In Stage 1, the model

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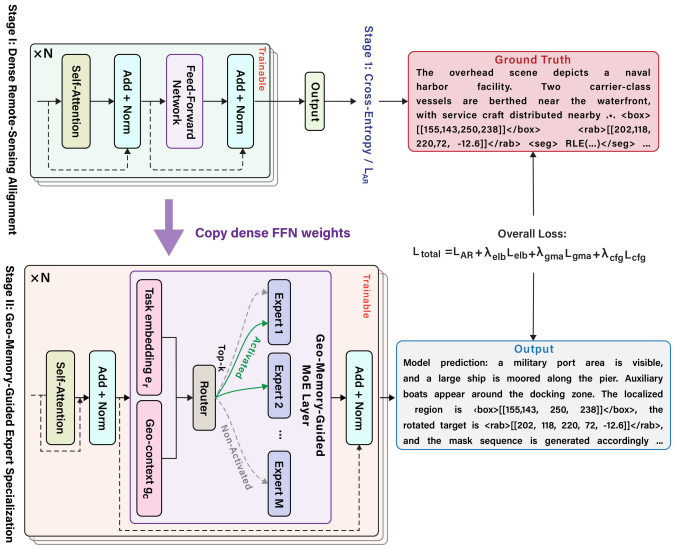


Fig. 4. Progressive dense-to-sparse training of GeoMEx-VLM. Stage 1 learns remote sensing visual–language alignment by optimizing the visual projection module and attention LoRA adapters while keeping the pretrained dense FFNs frozen. Stage 2 initializes sparse experts from the unchanged pretrained FFNs and optimizes the expert FFNs, routing parameters, and geo-memory components together with the projection module and attention LoRA adapters.

retains the dense architecture of the pretrained visual encoder and Vicuna decoder specified in Section II-A. Low-rank adaptation (LoRA) is applied to the query, key, value, and output projection matrices of every LLM self-attention layer, with rank $r = 64$ and scaling parameter $\alpha = 128$. The visual projection module and attention LoRA parameters are optimized, whereas the pretrained visual encoder and all original Vicuna parameters, including the dense FFN parameters, remain frozen. Therefore, Stage 1 establishes remote sensing visual–language alignment through projection learning and attention adaptation without updating the dense FFN weights.

In Stage 2, the dense FFN blocks at transformer depths $\{8, 16, 24\}$ are replaced with geo-memory-guided sparse MoE layers. Each expert is initialized by copying the corresponding pretrained Vicuna FFN parameters retained unchanged in the Stage 1 checkpoint. These parameters serve as initialization values rather than FFN weights learned during Stage 1. The visual projection module and attention LoRA parameters are carried forward from the Stage 1 checkpoint and continue to be optimized. The newly instantiated expert FFNs and routing parameters are trainable, whereas the retained original Vicuna parameters, including the dense FFNs at unreplaced layers, remain frozen. The CSGM module is activated to retrieve class- and scale-aware geo-context, refine the visual sequence, and condition top-k expert routing. Multiformat supervision supports AAB grounding, RAB grounding, segmentation, counting, scene understanding, VQA, captioning, and strategic asset grounding, enabling task- and granularity-specific expert specialization.

D. Training Objectives

GeoMEx-VLM is optimized with a compact multiobjective loss that jointly supervises semantic generation, spatial-token prediction, geo-memory learning, sparse expert routing, and cross-format grounding without task-specific heads. Since scene labels, captions, VQA answers, counts, AAB tokens, RAB tokens, and compressed mask tokens are represented as language-compatible sequences, autoregressive prediction serves as the main supervision. Auxiliary terms are used only to stabilize MoE routing, improve CSGM class discrimination, and enforce consistency among AAB, RAB, and mask representations. The overall training loss is defined as

$$\mathcal{L}_{total} = \mathcal{L}_{AR} + \lambda_{elb}\mathcal{L}_{elb} + \lambda_{gma}\mathcal{L}_{gma} + \lambda_{cfg}\mathcal{L}_{cfg} \tag{7}$$

where $\mathcal{L}_{AR}$ denotes the autoregressive generation loss, $\mathcal{L}_{elb}$ denotes the expert load-balancing loss, $\mathcal{L}_{gma}$ denotes the geo-memory alignment loss, and $\mathcal{L}_{cfg}$ denotes the cross-format grounding loss. The coefficients λ_{elb}, λ_{gma}, and λ_{cfg} control the contribution of the auxiliary terms.

Given image tokens V, geo-memory tokens V_{gmem}, instruction tokens Q, and a target sequence $Y = [y_1, y_2, \ldots, y_T]$, the autoregressive generation loss is formulated as

$$\mathcal{L}_{AR} = -\sum_{i=1}^{T} \log p_\theta \left(y_i \mid V, V_{gmem}, Q, y_{<i}\right). \tag{8}$$

This objective supervises both natural-language and spatial-token outputs, enabling GeoMEx-VLM to learn scene classification, VQA, captioning, counting, AAB grounding, RAB localization, and compressed mask generation without task-specific prediction heads.

To prevent expert collapse in sparse MoE layers, we use an expert load-balancing loss. Let E denote the number of experts, F_i denote the fraction of tokens routed to expert i, and G_i denote the average soft routing probability assigned to expert i within a mini-batch. The loss is defined as

$$\mathcal{L}_{elb} = E \sum_{i=1}^{E} F_i G_i. \tag{9}$$

This term encourages stable expert utilization while still allowing specialization according to task type, object category, scale, and spatial granularity.

For geo-memory supervision, the pooled CSGM representation is aligned with class prototypes. Let g_c denote the pooled geo-memory vector for class c, and let p_c denote its corresponding learnable prototype. The geo-memory alignment loss is formulated as

$$\mathcal{L}_{gma} = -\log \frac{\exp\left(\text{sim}\left(g_c, p_c\right)/\tau\right)}{\sum_{j=1}^{K} \exp\left(\text{sim}\left(g_c, p_j\right)/\tau\right)} \tag{10}$$

where $\text{sim}(\cdot,\cdot)$ denotes cosine similarity, τ is the temperature parameter, and K is the number of semantic classes. This term aligns retrieved geo-memory features with their correct class prototypes while separating them from other classes, strengthening CSGM for recurring categories such as aircraft, ships, runways, storage tanks, bridges, and dockyard structures. It is omitted when class-level grounding labels are unavailable.

Since the training datasets do not jointly provide native AAB, RAB, and mask annotations for the same instances, cross-format supervision is applied only to valid same-target pairs obtained through deterministic geometric conversion. For

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DOTA-v2.0 and FAIR1M, each native RAB b_o^N is paired with its enclosing AAB $b_h^D = f_{\text{rab}\to\text{aab}}(b_o^N)$, yielding AAB→RAB and RAB→AAB supervision. For iSAID and RefSegRS, each native mask M^N is paired with its tight enclosing AAB $b_h^D = f_{\text{mask}\to\text{aab}}(M^N)$, yielding mask→AAB supervision. Here, N and D denote native and deterministically derived annotations, respectively. No RAB→mask pseudo-labels or cross-dataset instance pairs are used. Accordingly, the set of valid cross-format pairs is

$$\mathcal{P} = \left\{ \left(t^D_{\text{aab}}, t^N_{\text{rab}}\right), \left(t^N_{\text{rab}}, t^D_{\text{aab}}\right), \left(t^N_{\text{mask}}, t^D_{\text{aab}}\right) \right\} \tag{11}$$

where each pair corresponds to the same annotated object or referred target. The cross-format grounding loss is then defined as

$$\mathcal{L}_{cfg} = -\frac{1}{|\mathcal{P}|} \sum_{(t^{src}, t^{tgt})\in\mathcal{P}} \sum_{i=1}^{T_g} \log p_\theta \left(t_i^{tgt} \mid I, Q, t^{src}, t_{<i}^{tgt}\right). \tag{12}$$

Thus, native annotations remain the primary supervision, while derived AABs serve only as auxiliary geometric bridges between spatial formats. Samples without a valid same-target conversion are excluded from $\mathcal{L}_{cfg}$ but remain supervised by the standard autoregressive objective.

E. Cross-Representation Spatial Agreement

For the same referred object, AAB, RAB, and compressed mask predictions should describe spatially consistent regions. However, because GeoMEx-VLM generates these outputs as decoded spatial tokens, IoU-based consistency over parsed coordinates is not directly differentiable through autoregressive decoding. Therefore, we use spatial agreement as an evaluation and diagnostic measure rather than as a training loss.

Let s_{aab}, s_{rab}, and s_{mask} denote the decoded AAB, RAB, and mask predictions, respectively. For compactness, define $b_r = f_{rab\to aab}(s_{rab})$ and $b_m = f_{mask\to aab}(s_{mask})$ as the enclosing AABs of the RAB and mask predictions. The spatial agreement score (SAS) is

$$\text{SAS} = \frac{\text{IoU}\left(s_{aab}, b_r\right) + \text{IoU}\left(s_{aab}, b_m\right) + \text{IoU}\left(b_r, b_m\right)}{3}. \tag{13}$$

A higher SAS indicates stronger agreement among AAB, RAB, and mask predictions. This diagnostic is particularly useful for small, rotated, elongated, or densely distributed objects, where predictions may be semantically correct but spatially inconsistent.

III. Experiment and Results

A. Dataset

To evaluate the semantic and spatial grounding capabilities of GeoMEx-VLM, we use diverse public remote sensing datasets covering instruction following, language-conditioned grounding, horizontal localization, rotation-aware localization, mask prediction, dense-object counting, and CSGM learning. These benchmarks contain several object categories relevant to asset monitoring, including aircraft, ships, harbors, airports, bridges, vehicles, storage tanks, and other infrastructure. DIOR/DIOR-RSVG [20], [21] supports AAB grounding, object counting, and text-guided asset localization; DOTA-v2.0 [22] provides oriented annotations for RAB grounding under large scale, shape, and orientation variation; FAIR1M [19] supports fine-grained strategic asset recognition and CSGM learning; iSAID [23] provides instance masks for compressed <seg> token learning; GeoChat-Instruct [1] is used for Stage-I dense vision–language alignment; RefSegRS [24] supports text-conditioned mask grounding; and xView [25] serves as a supplementary dataset for dense small-object localization, counting, and cross-scene generalization. Datasets are summarized in Supplementary Table S2. The datasets retain their native annotation modalities during primary supervision and evaluation. Cross-format supervision does not require a dataset to contain all spatial formats simultaneously: an enclosing AAB is deterministically computed from each native RAB in DOTA-v2.0 and FAIR1M and from each native target mask in iSAID and RefSegRS. These same-target conversions are used only by the auxiliary cross-format grounding objective; no RAB–mask annotation is assumed, and no spatial annotations are paired across datasets.

B. Evaluation Metrics

GeoMEx-VLM is evaluated according to each dataset's native supervision and the output format generated by Unified Spatial Tokenization. For DIOR-RSVG AAB grounding, we report Pr@0.5, Pr@0.7, Pr@0.9, mIoU, and cIoU; for DIOR and xView horizontal localization, AP@50, AP@75, and mAP; for DOTA-v2.0 and FAIR1M rotation-aware localization, rotated AP@50, AP@75, and rotated mAP; and for iSAID and RefSegRS mask-level grounding, mask AP, AP@50, AP@75, and mask mIoU. Object counting is evaluated using MAE, RMSE, and count accuracy, while VQA and scene classification use accuracy and captioning uses CIDEr, BLEU-4, METEOR, and ROUGE-L. We further report SAS to measure consistency among decoded AAB, RAB, and mask predictions after conversion to enclosing AABs, and sparse MoE expert-utilization statistics, including average expert load, routing entropy, and expert activation frequency. For datasets evaluated through high-resolution tiling, all predicted boxes, oriented boxes, masks, and object counts are first restored to the native image coordinate system and consolidated across overlapping tiles; the reported AP, mAP, IoU, and counting metrics are then computed at the original-image level rather than independently at the 336 $\times$ 336 tile level.

C. Experimental Setup

GeoMEx-VLM follows a two-stage dense-to-sparse schedule. Stage 1 optimizes the visual projection module and attention LoRA adapters while retaining frozen pretrained Vicuna FFNs. Stage 2 replaces selected dense FFN blocks with sparse MoE layers, initializes each expert from the corresponding unchanged pretrained FFN parameters in the Stage 1 checkpoint, and trains the expert FFNs, routing parameters, and CSGM components together with the projection module and attention LoRA adapters; the retained original Vicuna parameters remain frozen. All experiments are conducted on six NVIDIA A800-80G GPUs for five epochs with a global batch size of 144, using AdamW [26], an initial learning

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TABLE I

PERFORMANCE COMPARISON WITH RECENT STATE-OF-THE-ART RS-VLMS, GROUNDING MODELS, AND SEGMENTATION-ORIENTED REMOTE SENSING MLLMS. AAB-G DENOTES DIOR-RSVG PR@0.5, RAB-G DENOTES DOTA-V2.0 ROTATED MAP, RES DENOTES REFSEGRS MIOU, AND COUNT DENOTES COUNTING MAE ON THE DIOR/XVIEW COUNTING SUBSET. HIGHER IS BETTER EXCEPT MAE

Method	Model Type	VQA Acc.	CIDEr	AAB-G	RAB-G	RES	Count MAE
LLaVA-1.5 [27]	General dense MLLM	61.84	62.7	42.36	21.48	31.92	4.71
Qwen2.5-VL [28]	General dense MLLM	66.92	68.4	50.27	27.96	37.45	4.08
GeoChat [1]	Grounded RS-VLM	75.31	86.6	64.73	45.26	47.58	3.16
EarthGPT [5]	Multi-sensor RS-MLLM	77.49	91.4	66.15	48.74	49.83	2.94
RS-MoE [8]	Sparse MoE RS-MLLM	76.84	89.2	–	–	–	2.87
RSUniVLM [9]	Sparse MoE RS-MLLM	77.12	90.6	–	–	–	2.79
RingMoE [10]	Sparse MoE RS-MLLM	78.46	91.3	68.27	52.14	51.92	2.31
SkyMoE [11]	Sparse MoE RS-MLLM	79.31	92.5	70.48	54.37	54.06	2.18
RSThinker [41]	Geo-CoT SFT + GRPO RS-MLLM	77.24	93.8	**89.10**	–	–	–
GeoSolver [42]	Process-Aware Tree-GRPO RS-MLLM	73.19	95.2	87.80	–	–	–
GeoSearcher [43]	Anchor-centric SFT + PF-GRPO MLLM	74.91	–	83.20	57.33	72.58	–
GeoGround [13]	Unified grounding VLM	76.88	89.7	74.62	**60.41**	66.35	2.41
GeoPix [14]	Pixel-grounded RS-MLLM	79.74	96.2	73.19	56.88	69.81	2.46
GeoViS [31]	Visual-search grounding	78.95	92.4	79.80	55.26	54.91	2.37
GeoMEx-VLM	**Geo-memory sparse RS-VLM**	**84.92**	**112.7**	82.36	58.72	**75.42**	**1.74**

rate of 2×10^{-5}, and cosine learning-rate decay. Training is implemented in PyTorch with DeepSpeed ZeRO-2 data parallelism. The AdamW optimizer [26] is configured with $\beta_1 = 0.9$, $\beta_2 = 0.999$, and weight decay 1×10^{-2}. We employ a cosine learning-rate schedule with a 5% warm-up fraction, gradient clipping (max norm 1.0), and automatic mixed precision (BF16). Stage 1 runs for three epochs; Stage 2 runs for two epochs, yielding the five-epoch total reported above. The complete training process requires approximately 180 GPU-hours on 6× NVIDIA A800-80G GPUs.

We compare GeoMEx-VLM against dense RS-VLMs, sparse expert RS-VLMs, and task-specific grounding or segmentation models, and perform ablations by removing CSGM, replacing geo-memory-guided routing with token-only routing, disabling cross-format grounding loss, removing expert load balancing, and evaluating AAB-only, RAB-only, and mask-only tokenization. For all reported evaluations, CSGM operates without oracle annotations. Memory queries are inferred from the image–instruction pair, and any required spatial prior is model-generated. Ground-truth categories, scales, boxes, and masks are used only for training supervision and metric computation after prediction, not for memory retrieval or expert routing.

D. Performance Comparison

We evaluate GeoMEx-VLM against recent general-domain MLLMs, RS-VLMs, unified grounding models, pixel-grounded RS-MLLMs, visual-search and oriented-localization methods, referring segmentation approaches, and reasoning-oriented RS-MLLMs employing advanced post-training. The latter include RSThinker, which combines Geo-CoT supervised fine-tuning with GRPO; GeoSolver adopts process-aware Tree-GRPO and applies anchor-centric supervised fine-tuning followed by PF-GRPO. The comparison covers five evaluation dimensions: multitask RS-VL performance, language-conditioned AAB grounding, rotation-aware visual grounding, mask-level grounding and referring segmentation, and instruction-following with object counting and spatial agreement.

1) Multitask Performance: Table I shows that general-domain MLLMs, including LLaVA-1.5 [27] and Qwen2.5-VL [28], exhibit limited transfer to remote sensing spatial tasks, whereas RS-specific models such as GeoChat [1] and EarthGPT [5] provide substantial improvements. Among sparse MoE models, SkyMoE [11] outperforms RS-MoE [8], RSUniVLM [9], and RingMoE [10], achieving 79.31 VQA accuracy and 2.18 Count MAE. Reasoning-oriented post-training yields strong axis-aligned grounding performance, with RSThinker [41], GeoSolver [42], and GeoSearcher [43] obtaining AAB-G scores of 89.10, 87.80, and 83.20, respectively, although several task metrics are unavailable. Among spatially specialized models, GeoPix [14] achieves the CIDEr score (96.2), while GeoGround [13] records the highest RAB-G score (60.41). GeoMEx-VLM achieves the highest VQA accuracy (84.92), RES (75.42), lowest Count MAE (1.74), CIDEr (112.7), AAB-G (82.36), and RAB-G (58.72). In particular, it exceeds GeoSearcher [43] by 2.84 points in RES and reduces Count MAE by 0.44 relative to SkyMoE [11], demonstrating consistent performance across language, grounding, segmentation, and counting tasks.

2) Language-Conditioned Object Localization: Table II reports language-conditioned AAB grounding results on DIOR-RSVG. Among conventional RSVG methods, ProVG [35] achieves 81.64 Pr@0.5, 70.85 Pr@0.7, 39.16 Pr@0.9, 72.14 mIoU, and 82.05 cIoU, outperforming MGVLF [21], LPVA [33], and LGFormer [34] on most metrics. GeoViS [31] records the highest Pr@0.7 (73.42), while the reasoning-oriented GeoSolver [42] and GeoSearcher [43] achieve the highest mIoU (75.62) and 73.50 mIoU, respectively, although several metrics are unavailable. GeoMEx-VLM obtains the highest Pr@0.5 (82.36), Pr@0.9 (39.84), and cIoU (83.76), together with 68.91 Pr@0.7 and 70.91 mIoU. Compared with the strongest sparse MoE baseline, SkyMoE [11], GeoMEx-VLM improves Pr@0.5, Pr@0.9, and cIoU by 9.52, 6.42, and

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TABLE II

COMPARISON ON DIOR-RSVG LANGUAGE-CONDITIONED AAB GROUNDING. PR@t DENOTES PRECISION AT IOU THRESHOLD t

Method	Pr@0.5	Pr@0.7	Pr@0.9	mIoU	cIoU
MGVLF [21]	76.36	64.92	31.45	66.87	78.36
LPVA [33]	78.24	66.35	34.06	68.49	79.58
LGFormer [34]	80.27	69.14	37.52	71.08	81.32
ProVG [35]	81.64	70.85	39.16	72.14	82.05
GeoChat [1]	64.73	50.84	19.62	58.43	67.91
RingMoE [10]	70.31	60.72	31.85	63.27	72.15
SkyMoE [11]	72.84	62.91	33.42	65.10	74.08
GeoGround [13]	74.62	61.78	30.25	66.54	75.86
GeoViS [31]	79.80	**73.42**	36.75	70.63	80.27
GeoSolver† [42]	79.80	–	–	**75.62**	–
GeoSearcher† [43]	81.20	72.40	–	73.50	–
GeoMEx-VLM	**82.36**	68.91	**39.84**	70.91	**83.76**

TABLE III

INSTRUCTION-FOLLOWING, COUNTING, AND SPATIAL AGREEMENT COMPARISON. COUNT ACCURACY DENOTES EXACT-MATCH COUNT PREDICTION. SAS MEASURES AGREEMENT AMONG DECODED AAB, RAB, AND MASK PREDICTIONS AFTER CONVERSION TO ENCLOSING AABS

Method	VQA	Scene	CIDEr	BLEU	MAE	C-Acc.	SAS
LLaVA-1.5 [27]	61.84	70.23	62.7	18.42	4.71	42.36	41.28
Qwen2.5-VL [28]	66.92	74.85	68.4	20.36	4.08	47.19	44.75
GeoChat [1]	75.31	82.74	86.6	27.43	3.16	57.84	56.31
EarthGPT [5]	77.49	84.18	91.4	29.05	2.94	60.42	58.67
GeoGround [13]	76.88	83.61	89.7	28.16	2.41	66.93	68.74
EarthGPT-X [7]	81.32	87.54	101.5	32.17	2.18	70.26	72.81
RS-MoE [8]	76.84	84.02	89.2	28.75	2.87	65.24	–
RSUniVLM [9]	77.12	84.55	90.6	29.31	2.79	66.10	–
RingMoE [10]	78.46	85.73	91.3	30.24	2.31	68.42	61.38
SkyMoE [11]	79.31	86.48	92.5	31.02	2.18	69.87	63.26
GeoLLaVA-8K [29]	82.07	88.63	103.1	**36.94**	2.06	71.54	71.69
Earth-OneVision [30]	83.46	89.62	109.8	34.82	1.96	73.48	**74.35**
GeoMEx-VLM	**84.92**	**90.47**	**112.7**	32.08	**1.74**	**76.35**	74.02

9.68 percentage points, respectively, demonstrating improved strict localization accuracy and overall spatial consistency.

3) Object Counting and Spatial Agreement: Table III compares instruction-following, scene classification, captioning, counting, and spatial agreement. LLaVA-1.5 [27] and Qwen2.5-VL [28] show limited transfer (VQA 61.84/66.92, MAE 4.71/4.08, and SAS 41.28/44.75). Dense RS-VLMs improve: GeoChat [1] reaches 75.31 VQA and 3.16 MAE; EarthGPT [5] achieves 77.49 VQA, 91.4 CIDEr, and 2.94 MAE; GeoGround [13] obtains 2.41 MAE and 68.74 SAS; and EarthGPT-X [7] reaches 81.32 VQA, 2.18 MAE, and 72.81 SAS. Among sparse MoE baselines, SkyMoE [11] performs best (79.31 VQA, 92.5 CIDEr, 2.18 MAE, and 63.26 SAS), followed by RingMoE [10] (78.46 VQA, 2.31 MAE, and 61.38 SAS); RS-MoE [8] and RSUniVLM [9] are limited to language and counting tasks, so SAS is not reported. GeoLLaVA-8K [29] achieves the best BLEU-4 (36.94), and Earth-OneVision [30] achieves the best baseline SAS (74.35). GeoMEx-VLM obtains the highest VQA (84.92), scene accuracy (90.47), CIDEr (112.7), and count accuracy (76.35), with the lowest MAE (1.74) and a competitive SAS of 74.02, outperforming SkyMoE by 1.61 VQA, 6.48 count accuracy, and 10.76 SAS while reducing MAE by 0.44.

E. Ablation Study

1) Sensitivity to Geo-Memory Bank Capacity: Table IV evaluates the GMB capacity N_g and the number of scale bins

TABLE IV

ABLATION ON GMB CAPACITY. N_g DENOTES MEMORY SLOTS PER CLASS–SCALE PAIR; L DENOTES THE NUMBER OF SCALE BINS. DEFAULT SETTINGS ARE $N_g = 16$ AND $L = 4$

Configuration	AAB-G	RAB-G	RES	Count MAE	SAS
$N_g = 2,\ L = 4$	76.42	54.18	70.15	2.38	68.45
$N_g = 4,\ L = 4$	77.85	56.31	72.48	2.05	71.20
$N_g = 8,\ L = 4$	79.12	57.64	74.06	**1.71**	72.83
$N_g = 16,\ L = 4$	**82.36**	58.72	**75.42**	1.74	**74.02**
$N_g = 32,\ L = 4$	82.18	**59.15**	75.28	1.82	73.45
$N_g = 16,\ L = 2$	78.94	55.63	71.82	2.01	70.56
$N_g = 16,\ L = 3$	80.75	57.89	74.35	1.82	73.14
$N_g = 16,\ L = 5$	81.84	58.41	75.06	1.79	73.68
$N_g = 16,\ L = 6$	81.56	58.12	74.71	1.81	73.45

TABLE V

SENSITIVITY TO EXPERT CARDINALITY E AND TOP-k ROUTING. DEFAULT: $E = 8$, $k = 2$. AVG. LOAD DENOTES THE MEAN FRACTION OF EXPERTS ACTIVATED PER LAYER; ROUTING ENTROPY MEASURES THE UNIFORMITY OF THE ROUTING DISTRIBUTION

E	k	VQA	AAB-G	RAB-G	RES	Count MAE	Avg. Load	Routing Ent.
4	1	77.42	74.15	54.82	70.36	2.24	0.25	0.62
4	2	78.18	75.63	56.45	71.95	2.08	0.50	0.71
8	1	79.05	76.28	56.91	72.84	1.95	0.13	0.58
8	2	**84.92**	**82.36**	**58.72**	75.42	**1.74**	0.25	0.74
8	4	80.56	77.95	58.31	**75.68**	1.81	0.50	0.82
16	2	81.05	78.12	58.45	75.18	1.78	0.13	0.76
16	4	80.64	77.84	58.19	74.76	1.83	0.25	0.85
32	2	80.71	77.93	58.28	74.95	1.80	0.06	0.72

L. Small N_g values reduce RAB grounding and mask IoU because limited prototypes cannot capture sufficient intraclass variation, while gains saturate beyond $N_g = 16$. Although $N_g = 32$ slightly improves RAB-G, it increases retrieval ambiguity and degrades Count MAE and SAS. In contrast, $N_g = 8$ achieves the best Count MAE by reducing retrieval noise in dense scenes. For scale quantization, $L = 4$ provides the best balance; $L = 2$ is too coarse, whereas $L = 6$ overpartitions the memory and sparsely populates scale bins.

2) Expert Cardinality and Top-k Sparsity: Table V analyzes the effects of expert count E and top-k routing. Increasing E from 4 to 8 consistently improves spatial performance by enabling stronger expert specialization, whereas larger expert sets provide only marginal language gains and slightly reduce spatial efficiency. Top-k routing performs best at $k = 2$: $k = 1$ is overly restrictive for compositional reasoning, while $k = 4$ increases activation density and slightly improves RES at the cost of specialization and counting accuracy. Therefore, $E = 8$ and $k = 2$ provide the best overall accuracy–efficiency tradeoff, activating only 25% of the expert FFN parameters per token.

3) Impact of Instruction Corpus and Geo-Context Augmentation: Table VI evaluates the instruction construction and geo-context augmentation pipeline. The full configuration achieves the best overall FAIR1M rare-class r-mAP, DOTA-v2.0 r-mAP, and SAS, confirming the benefit of class-scale sampling, cross-format instruction expansion, and geo-context augmentation. Removing geometric remapping mainly degrades rotated grounding, while removing cross-format expansion reduces SAS. Disabling memory-balanced sampling weakens rare-class performance. Although removing context-contrastive composition slightly improves xView mAP, it reduces FAIR1M performance, indicating a tradeoff between dense-object localization and contextual robustness.

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TABLE VI

ABLATION STUDY ON INSTRUCTION CONSTRUCTION AND GEO-CONTEXT AUGMENTATION. XVIEW REPORTS MAP FOR DENSE SMALL-OBJECT ROBUSTNESS, FAIR1M REPORTS RARE-CLASS ROTATED MAP, DOTA-V2.0 REPORTS ROTATED MAP, AND SAS MEASURES CROSS-FORMAT SPATIAL AGREEMENT

Variant	xView	FAIR1M	DOTA	SAS
Native annotations	64.20	42.15	56.40	62.30
w/o class-scale sampling	67.50	46.20	64.10	73.10
w/o instruction expansion	66.90	52.45	63.80	69.15
w/o asset composition	**68.85**	49.60	63.15	72.40
w/o geometric remapping	67.80	51.15	58.20	71.30
w/o geo-context aug	65.15	48.30	61.50	68.45
Instruction corpus + geo-context aug	68.42	**54.20**	**64.72**	**74.02**

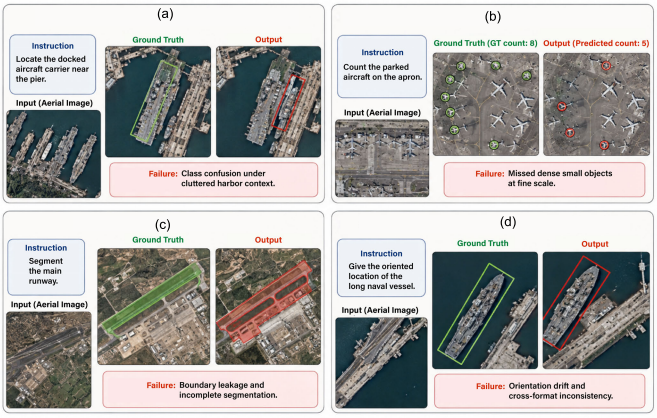


Fig. 5. Representative failure cases of GeoMEx-VLM, including harbor class confusion, dense aircraft undercounting, runway boundary leakage, and vessel orientation drift. (a) Harbor—class confusion in visual grounding. (b) Airport apron—dense small-object counting failure. (c) Airbase—segmentation failure. (d) Naval scene—rotating-aware ground failure.

F. Failure Case Analysis

Fig. 5 illustrates representative failure cases of GeoMEx-VLM under complex RS conditions. In harbor scenes, the model may confuse visually similar naval structures and localize an adjacent vessel instead of the queried docked aircraft carrier, indicating residual class ambiguity under cluttered maritime context. In dense airport-apron scenes, GeoMEx-VLM undercounts parked aircraft when small instances are closely spaced, partially occluded, or visually blended with apron markings and shadows. For runway segmentation, the predicted mask leaks into nearby taxiways and apron regions, showing that elongated infrastructure with similar texture and geometry remains challenging for compressed mask-token decoding. In rotation-aware grounding, the model localizes the naval vessel but exhibits orientation drift and spatial offset, revealing that RAB prediction can still be sensitive to elongated targets, nearby distractors, and imperfect cross-format geometric consistency.

IV. CONCLUSION

This article presented GeoMEx-VLM for multigranularity RS interpretation and strategic asset grounding. The model integrates CSGM, geo-memory-guided expert routing, and unified spatial tokenization to support scene understanding, VQA, captioning, counting, AAB grounding, RAB localization, and compressed mask prediction within one autoregressive framework. By injecting class- and scale-conditioned geospatial priors into both visual modeling and sparse routing, GeoMEx-VLM improves the recognition and localization of small, rotated, dense, and visually similar remote sensing objects. Experiments demonstrate strong semantic and spatial performance. GeoMEx-VLM achieves 84.92 VQA accuracy, 112.7 CIDEr, 82.36 AAB-G, 58.72 RAB-G, 75.42 RES, and the lowest Count MAE of 1.74 in the overall comparison. It further obtains 82.36 Pr@0.5 and 83.76 cIoU on DIOR-RSVG, 64.72 r-mAP on DOTA-v2.0, 79.91 rAP_{50} on FAIR1M, and 85.04 Pr@0.5 with 82.16 oIoU on RefSegRS. Despite these gains, GeoMEx-VLM remains challenged by highly cluttered scenes, dense small-object counting, elongated-structure segmentation, and orientation-sensitive localization when similar targets appear close together. Future work will explore higher resolution visual tokenization, stronger instance-level disambiguation, geometry-constrained decoding, and uncertainty-aware counting for more robust strategic RS interpretation.

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